

Project Title:**Identifying High-Potential but Under-Engaged Customers Through Education, Income, Campaign Behavior, and Macroeconomic Context****Research Question**

Among highly educated customers (PhD, Master's), spending is highest yet campaign responsiveness is lowest. This project investigates whether education level, relative income, and the macroeconomic environment at enrollment (CPI) can explain this paradox and help identify high-potential but under-engaged customer segments.

This project investigates a behavioral paradox observed in customer data: highly educated customers — particularly those with PhD-level education — exhibit the highest average spending across all product categories, yet 67.5% have never responded to any of the six marketing campaigns in the dataset. The central research question is: Can education level, relative income positioning, campaign response history, and the macroeconomic environment at enrollment (CPI) jointly identify high-potential but under-engaged customers — and what distinguishes those who are campaign-immune from similarly educated customers who do engage?

The primary dataset is the Customer Personality Analysis dataset (2,240 customers, 29 features), covering enrollments from January to December 2013. It includes demographic variables (education across 5 tiers, marital status, household composition), product-level spending (wines, fruits, meat, fish, sweets, gold), purchase channel data, and six binary campaign response indicators (AcceptedCmp1–5 and Response), which will be combined into a composite Campaign Engagement Score. This is enriched with monthly US CPI data (918 observations, 1947–2023), matched to each customer's enrollment month to capture inflation context, and an external salary benchmark by education level to construct a relative income measure.

The analysis will include data preprocessing, RFM feature engineering, CPI cohort tagging, and relative income construction, followed by K-Means clustering evaluated via the Elbow Method and Silhouette Score. The expected outcome is a data-driven characterization of the 'campaign-immune but high-potential' segment and actionable targeting recommendations for reaching customers who are unreachable through standard campaigns.